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United States Patent	12406840
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Wang; Peimeng

Semiconductor structure and forming method thereof

Abstract

The present application provides a semiconductor structure and a forming method thereof. The method of forming the semiconductor structure includes: forming a capacitor base, the capacitor base including a plurality of capacitor switching structures and an isolation layer located between adjacent capacitor switching structures and covering top surfaces of the capacitor switching structures; removing the isolation layer covering the top surfaces of the capacitor switching structures, and exposing the capacitor switching structures; oxidizing a surface of the capacitor base exposing the capacitor switching structures, and forming an oxide layer; and removing the oxide layer, and exposing the capacitor switching structures.

Inventors:	Wang; Peimeng (Hefei, CN)
Applicant:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
Family ID:	85152644
Assignee:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
Appl. No.:	17/660060
Filed:	April 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230041544 A1	Feb. 09, 2023

Foreign Application Priority Data

CN	202110890950.6	Aug. 04, 2021
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Related U.S. Application Data

Publication Classification

Int. Cl.: H01L21/02 (20060101); H01L21/3105 (20060101); H10B12/00 (20230101)

U.S. Cl.:

CPC H01L21/02065 (20130101); H01L21/31055 (20130101); H10B12/03 (20230201)

Field of Classification Search

CPC: H10B (12/03); H10B (12/0335); H01L (21/02244); H01L (21/31055); H01L (21/02252); H01L (21/02065); H01L (21/02068); H01L (21/31116)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/CN2022/076281, filed on Feb. 15, 2022, which claims the priority of Chinese Patent Application No. 202110890950.6, filed to the CNIPA on Aug. 4, 2021 and titled “SEMICONDUCTOR STRUCTURE AND FORMING METHOD THEREOF”. The entire contents of International Application No. PCT/CN2022/076281 and Chinese Patent Application No. 202110890950.6 are incorporated herein by reference.

TECHNICAL FIELD

(1) The present application relates to the technical field of integrated circuits (ICs), in particular to a semiconductor structure and a forming method thereof.

BACKGROUND

(2) Dynamic random access memory (DRAM) is a commonly used semiconductor structure in electronic devices such as computers. It is composed of multiple memory cells, each of which usually includes a capacitor for storage of charges and a transistor for access of the capacitor. In the transistor, the gate is electrically connected to a word line (WL), the source is electrically connected to a bit line (BL), and the drain is electrically connected to the capacitor. The voltage on the WL controls the on or off of the transistor, so as to read data information stored in the capacitor or write data information into the capacitor through the BL.

(3) The capacitor is connected to the drain through capacitor switching structures in the capacitor base. However, due to restrictions of existing manufacturing processes, it is likely to cause defects on the surface of the capacitor base and short circuits between adjacent capacitor switching structures.

(4) Therefore, how to reduce the defects on the surface of the capacitor base and avoid the short circuits between the adjacent capacitor switching structures to improve electrical properties of the semiconductor structure is a technical problem to be solved urgently.

SUMMARY

(5) The present application provides a semiconductor structure and a forming method thereof.

(6) According to some embodiments, the present application provides a method of forming a semiconductor structure, including the following steps: forming a capacitor base, the capacitor base including a plurality of capacitor switching structures and an isolation layer located between adjacent capacitor switching structures and covering top surfaces of the capacitor switching structures; removing the isolation layer covering the top surfaces of the capacitor switching structures, and exposing the capacitor switching structures; oxidizing a surface of the capacitor base exposing the capacitor switching structures, and forming an oxide layer; and removing the oxide layer, and exposing the capacitor switching structures.

(7) According to other embodiments, the present application further provides the semiconductor structure, which is formed with the above the method of forming the semiconductor structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a flowchart of a method of forming a semiconductor structure according to a specific implementation of the present application;

(2) FIG. 2A to FIG. 2D each are a schematic cross-sectional view of a main process during

formation of a semiconductor structure according to a specific implementation of the present application;

- (3) FIG. 3A to FIG. 3D each are a schematic cross-sectional view of a main process during formation of an oxide layer according to a specific implementation of the present application; and
- (4) FIG. 4A to FIG. 4E each are a schematic cross-sectional view of a main process during removal of an oxide layer according to a specific implementation of the present application.

DETAILED DESCRIPTION

(5) Specific implementations of a semiconductor structure and a forming method thereof provided in the present application will be described below in detail with reference to the accompanying drawings.

(6) A specific implementation of the present application provides a method of forming a semiconductor structure. FIG. 1 is a flowchart of a method of forming a semiconductor device according to a specific implementation of the present application. FIG. 2A to FIG. 2D each are a schematic cross-sectional view of a main process during formation of a semiconductor structure according to a specific implementation of the present application. As shown in FIG. 1 and FIG. 2A to FIG. 2D, the method of forming a semiconductor structure includes the following steps: Step S11: Form a capacitor base **21**, the capacitor base **21** including a plurality of capacitor switching structures **211** and an isolation layer **212** located between adjacent capacitor switching structures **211** and covering top surfaces of the capacitor switching structures **211**, as shown in FIG. 2A.

(7) In some embodiments, the step of forming a capacitor base **21** specifically includes: Provide a substrate **20**, a plurality of capacitor contact regions being provided in the substrate **20**.

(8) Form the plurality of the capacitor switching structures **211** on a surface of the substrate **20**, the plurality of the capacitor switching structures **211** being electrically connected to the plurality of the capacitor contact regions in one-to-one correspondence.

(9) Form the isolation layer **212** filling gaps between the adjacent capacitor switching structures **211** and covering the top surfaces of the capacitor switching structures **211**.

(10) Specifically, the substrate **20** may be, but not limited to, a silicon substrate. The specific implementation is described by taking the substrate **20** as the silicon substrate for example. In other examples, the substrate **20** may be a semiconductor substrate such as a gallium nitride substrate, a gallium arsenide substrate, a gallium carbide substrate, a silicon carbide substrate or a silicon-on-insulator (SOI) substrate. A plurality of active regions are arranged in the substrate **20** in an array. The active regions each include a BL contact region and a capacitor contact region. Prior to formation of the plurality of capacitor switching structures **211**, a capacitor contact layer may further be formed on a surface of the substrate **20**. Capacitor contact points electrically contacting the plurality of capacitor contact regions in one-to-one correspondence are provided in the capacitor contact layer. The capacitor contact points may be made of polycrystalline silicon.

Thereafter, the plurality of capacitor switching structures **211** electrically connected to a plurality of the capacitor contact points in one-to-one correspondence are formed. The capacitor switching structures **211** may be made of a conductive metal material such as tungsten. The plurality of capacitor switching structures **211** are independent of each other, namely gaps are provided between adjacent capacitor switching structures **211**. In order to avoid influences between adjacent capacitor switching structures **211**, after formation of the plurality of capacitor switching structures **211**, an insulating material is deposited in the gaps between the adjacent capacitor switching structures **211** and covers the top surfaces of the capacitor switching structures **211** to form the isolation layer **212** shown in FIG. 2A. The isolation layer **212** may be made of an oxide material (such as silicon dioxide), and may also be made of a nitride material (such as silicon nitride). Step S12: Remove the isolation layer **212** covering the top surfaces of the capacitor switching structures **211**, and expose the capacitor switching structures **211**, as shown in FIG. 2B.

(11) In some embodiments, the step of removing the isolation layer **212** covering the top surfaces of the capacitor switching structures **211** specifically includes: Etch the isolation layer **212** with a

dry etching process.

(12) In some embodiments, the isolation layer **212** is made of a nitride material; and the step of etching the isolation layer **212** with a dry etching process specifically includes: Etch the isolation layer **212** with a mixed gas of CF₄, CHF₃ and O₂ as an etching gas.

(13) Specifically, when the isolation layer **212** is made of the nitride material such as silicon nitride, the mixed gas of CF₄, CHF₃ and O₂ may be used as the etching gas to etch the isolation layer **212** downward along a direction (namely a direction perpendicular to the surface of the capacitor base **21**) indicated by arrows in FIG. 2B. With the capacitor switching structures **211** as etching cutoff layers, a part of the isolation layer **212** is removed to expose the top surfaces of the capacitor switching structures **211**, as shown in FIG. 2B. When the mixed gas of CF₄, CHF₃ and O₂ is used as the etching gas for etching, plasmas formed by the mixed gas of CF₄, CHF₃ and O₂ will damage the top surfaces of the capacitor switching structures **211**. As a result, material particles in the capacitor switching structures **211** are bombarded, at least one of the material particles is further oxidized easily by O₂ to form particulate matters **22** including the material particles and material particle oxides, and the particulate matters **22** are adhered on the surface of the capacitor base **21**, as shown in FIG. 2B.

(14) With the capacitor switching structures **211** made of the tungsten as an example, the plasmas formed by the mixed gas of CF₄, CHF₃ and O₂ bombard the surfaces of the capacitor switching structures **211**, such that tungsten particles and tungsten oxide particles are left on the surface of the capacitor base **21**.
Step S13: Oxidize a surface of the capacitor base **21** exposing the capacitor switching structures **211**, and form an oxide layer **23**, as shown in FIG. 2C.

(15) In the specific implementation, upon exposure of the top surfaces of the capacitor switching structures **211**, the material particles left on the surface of the capacitor base **21** and the top surfaces of the capacitor switching structures **211** are oxidized to form the oxide layer **23** including same components as the material particle oxide. The uniform material on the top surface of the capacitor base **21** facilitates subsequent thorough removal of the particulate matters **22** on the surface of the capacitor base **21**, and avoids the short circuits between the adjacent capacitor switching structures **211**. By partially or completely removing damaged regions on the surfaces of the capacitor switching structures **211**, defects on the surfaces of the capacitor switching structures **211** are reduced or completely eliminated.

(16) In some embodiments, the step of oxidizing a surface of the capacitor base **21** exposing the capacitor switching structures **211** specifically includes: Oxidize, with an O₂ plasma, the surface of the capacitor base **21** exposing the capacitor switching structures **211**.

(17) In some embodiments, a reaction temperature is 25-300° C. when the O₂ plasma is used to oxidize the surface of the capacitor base **21** exposing the capacitor switching structures **211**. For example, when the O₂ plasma is used to oxidize the surface of the capacitor base **21** exposing the capacitor switching structures **211**, the reaction temperature is 25° C.-100° C., 150° C.-200° C. or 200° C.-250° C., preferably 200° C.-250° C.

(18) In some embodiments, the step of oxidizing, with an O₂ plasma, the surface of the capacitor base **21** exposing the capacitor switching structures **211** specifically includes: Transmit a mixed gas plasma including at least the O₂ plasma and an H₂N₂ plasma to the surface of the capacitor base **21** exposing the capacitor switching structures **211**.

(19) In some embodiments, the mixed gas plasma has a flow of 100 sccm-15,000 sccm. For example, the mixed gas plasma has the flow of 100 sccm-500 sccm, 400 sccm-100 sccm, 1,000 sccm-5,000 sccm, 8,000 sccm-12,000 sccm or 10,000 sccm-15,000 sccm, preferably 8,000 sccm-12,000 sccm. At an initial stage of oxidizing, with the O₂ plasma, the surface of the capacitor base **21** exposing the capacitor switching structures **211**, there may be a relatively high flow of the mixed gas plasma. At a nearly saturated stage of oxidization, the flow of the mixed gas plasma may be adjusted to be relatively low. The surfaces of the capacitor switching structures **211** are oxidized in case of the relatively high flow of the mixed gas plasma, and reaction residues are removed in

case of the relatively low flow of the mixed gas plasma.

(20) In some embodiments, the mixed gas plasma has a pressure of 10 mtorr-10,000 mtorr. For example, the mixed gas plasma has the pressure of 10 mtorr-100 mtorr, 50 mtorr-500 mtorr, 500 mtorr-1,200 mtorr or 2,000 mtorr-10,000 mtorr, preferably 500 mtorr-1,200 mtorr.

(21) In some embodiments, the step of transmitting a mixed gas plasma including at least the O.sub.2 plasma and an H.sub.2N.sub.2 plasma to the surface of the capacitor base **21** exposing the capacitor switching structures **211** specifically includes: Ionize, with an RF power of 100 W-10,000 W, a mixed gas including O.sub.2 and H.sub.2N.sub.2 to form the mixed gas plasma. For example, the RF power may be 100 W-500 W, 500 W-1,000 W, 2,500 W-5,000 W or 5,000 W-10,000 W, preferably 2,500 W-5,000 W. Transmit the mixed gas plasma to the surface of the capacitor base **21** exposing the capacitor switching structures **211**.

(22) FIG. 3A to FIG. 3D each are a schematic cross-sectional view of a main process during formation of an oxide layer according to a specific implementation of the present application. For example, the oxide layer **23** is specifically formed as follows: A reaction chamber **30** is provided, wherein the reaction chamber **30** is provided therein with a support table **31**, and a showerhead **32** on an upper part of the reaction chamber **30**; the showerhead communicates with an air inlet channel **33**; and before the capacitor base **21** is placed into the reaction chamber **30**, the support table **31** for supporting the capacitor base **21** is preheated to 25° C.-300° C., as shown in FIG. 3A. The capacitor base **21** is placed on a surface of the support table **31**, and a mixed gas of O.sub.2, H.sub.2N.sub.2 and N.sub.2 or a mixed gas of O.sub.2, H.sub.2N.sub.2 and Ar as a wet etching gas is transmitted to the reaction chamber **30** through the air inlet channel **33**, wherein the dry etching gas is uniformly dispersed by the showerhead **32** and ionized at an RF power of 100 W-10,000 W to form a mixed gas plasma including an O.sub.2 plasma, an H.sub.2N.sub.2 plasma and an N.sub.2 plasma or a mixed gas plasma including an O.sub.2 plasma, an H.sub.2N.sub.2 plasma and an Ar plasma, and the mixed gas plasma oxidizes the surface of the capacitor base **21** to form the oxide layer **23**, as shown in FIG. 3B. In the step herein, the mixed gas plasma has a flow of 100 sccm-15,000 sccm and a pressure of 10 mtorr-10,000 mtorr. With the dry etching gas at the low flow and low pressure, the oxidation endpoint is controlled conveniently to avoid excessive oxidation of the capacitor switching structures **211**. N.sub.2 purging is continuously carried out on the reaction chamber **30** to cool the capacitor base **21** to a room temperature, as shown in FIG. 3C. The capacitor base **21** is taken out from the reaction chamber **30**. Step S14: Remove the oxide layer **23**, and expose the capacitor switching structures **211**, as shown in FIG. 2D.

(23) In some embodiments, the step of removing the oxide layer **23** specifically includes: Clean the capacitor base **21**.

(24) In some embodiments, the step of cleaning the capacitor base **21** specifically includes: Clean the capacitor base **21** with a DHF solution.

(25) In some embodiments, a volume ratio of HF to H.sub.2O in the DHF solution is (10:1)-(1,000:1). For example, the volume ratio of the HF to the H.sub.2O in the DHF solution is (10:1)-(50:1), (20:1)-(100:1), (200:1)-(800:1) or (500:1)-(1,000:1), preferably (200:1)-(800:1).

(26) FIG. 4A to FIG. 4E each are a schematic cross-sectional view of a main process during removal of an oxide layer according to a specific implementation of the present application. For example, the step of cleaning the capacitor base **21** specifically includes: Deionized water is sprayed to the oxide layer **23** on the capacitor base **21** through a first spray tube **40** to wet the oxide layer **23**, as shown in FIG. 4A. The DHF (HF diluted with the deionized water) solution is continuously sprayed to the surface of the oxide layer **23** of the capacitor base **21** through the first spray tube **40**, to remove particulate pollutants and native oxides on the surface of the oxide layer **23** and expose the oxide layer **23**, as shown in FIG. 4B.

(27) In some embodiments, the step of removing the oxide layer **23** specifically includes: Remove the oxide layer **23** with a wet etching process.

(28) In some embodiments, the step of removing the oxide layer **23** with a wet etching process

specifically includes: Remove the oxide layer **23** with a wet etchant having an etch selectivity of greater than or equal to 10:1 for the oxide layer **23** and the capacitor switching structures **211**.

(29) In the specific implementation, the particulate matters left on the surface of the capacitor base **21** contain uniform components by oxidation, and components in the oxide layer **23** are different from those in the capacitor switching structures **211**. With the etchant having the high etch selectivity for the oxide layer **23** and the capacitor switching structures **211**, the oxide layer **23** is fully removed, without damaging the capacitor switching structures **211**.

(30) In some embodiments, the capacitor switching structures **211** are made of a metal material, and the step of removing the oxide layer **23** with a wet etching process specifically includes: Remove the oxide layer **23** with an alkaline solution as a wet etchant.

(31) In some embodiments, the step of removing the oxide layer **23** with an alkaline solution as a wet etchant specifically includes: Remove the oxide layer **23** with a mixed solution of NH_4OH and H_2O as the wet etchant.

(32) In some embodiments, a volume ratio of the NH_4OH to the H_2O in the wet etchant is (5:1)-(1,000:1). For example, the volume ratio of the NH_4OH to the H_2O in the wet etchant is (5:1)-(100:1), (20:1)-(200:1), (50:1)-(500:1) or (100:1)-(1,000:1), preferably (50:1)-(500:1).

(33) In some embodiments, after removing the oxide layer **23**, the forming method further includes the following step: Dry the capacitor base **21**.

(34) In some embodiments, the step of drying the capacitor base **21** specifically includes: Purge the capacitor base **21** with a mixed gas of nitrogen and isopropanol.

(35) The following descriptions are made by taking the capacitor switching structures **211** made of the tungsten, and the oxide layer **23** made of the tungsten oxide as an example. An ADM solution including NH_4OH and H_2O is sprayed to the surface of the oxide layer **23** through the first spray tube **40**, as shown in FIG. 4C. The ADM solution can react with the tungsten oxide, rather than the tungsten. The ADM solution has an etch selectivity of 50:1 for the tungsten oxide and the tungsten, thus fully removing the tungsten oxide without damaging the tungsten. As shown in FIG. 4D, deionized water is sprayed to the surface of the capacitor base **21** again through the first spray tube **40** to remove the residual ADM solution. As shown in FIG. 4E, the mixed gas of nitrogen and isopropanol is sprayed to the surface of the capacitor base **21** by inclining the second spray tube **41** (for example, toward a direction indicated by the arrow in FIG. 4E) to avoid oxidation of the tungsten due to residual moisture.

(36) A specific implementation of the present application further provides a semiconductor structure, which is formed with the above method of forming a semiconductor structure.

(37) According to the semiconductor structure and the forming method thereof provided in the specific implementations, after the isolation layer is removed and the top surfaces of the capacitor switching structures are exposed, the surfaces of the capacitor switching structures and conductive particulate matters left on the surfaces of the capacitor switching structures are oxidized to implement the uniform surface material of the capacitor base. Upon the removal of the oxide layer, the defects on the surfaces of the capacitor switching structures as well as the particulate matters on the surfaces of the capacitor switching structures and on the surface of the isolation layer can be removed. Therefore, the present application reduces the defects on the surfaces of the capacitor switching structures and avoids the short circuits between the capacitor switching structures to improve electrical properties of the semiconductor structure.

(38) The above described are merely preferred implementations of the present application. It should be noted that several improvements and modifications may further be made by a person of ordinary skill in the art without departing from the principle of the present application, and such improvements and modifications should also be deemed as falling within the protection scope of the present application.

Claims

1. A method of forming a semiconductor structure, comprising: forming a capacitor base, the capacitor base comprising a plurality of capacitor switching structures and an isolation layer located between adjacent capacitor switching structures and covering top surfaces of the capacitor switching structures, wherein the isolation layer is made of a nitride material; etching the isolation layer with a mixed gas of CF_4 , CHF_3 and O_2 as an etching gas to expose the top surfaces of the capacitor switching structures, wherein particulate matters are adhered on the top surfaces of the capacitor switching structures and a surface of the isolation layer during etching the isolation layer; oxidizing, with an O_2 plasma, the top surfaces of the capacitor switching structures, the surface of the isolation layer and the particulate matters to form an oxide layer on the top surfaces of the capacitor switching structures and the surface of the isolation layer; and removing the oxide layer to expose the top surfaces of the capacitor switching structures.
2. The method of forming the semiconductor structure according to claim 1, wherein the forming a capacitor base comprises: providing a substrate, a plurality of capacitor contact regions being provided in the substrate; forming the plurality of the capacitor switching structures on a surface of the substrate, the plurality of the capacitor switching structures being electrically connected to the plurality of the capacitor contact regions in one-to-one correspondence; and forming the isolation layer filling gaps between the adjacent capacitor switching structures and covering the top surfaces of the capacitor switching structures.
3. The method of forming the semiconductor structure according to claim 1, wherein a reaction temperature is 25°C .- 300°C . when the O_2 plasma is used to oxidize the top surfaces of the capacitor switching structures, the surface of the isolation layer and the particulate matters.
4. The method of forming the semiconductor structure according to claim 1, wherein oxidizing, with an O_2 plasma, the top surfaces of the capacitor switching structures, the surface of the isolation layer and the particulate matters to form an oxide layer on the top surfaces of the capacitor switching structures and the surface of the isolation layer comprises: transmitting a mixed gas plasma comprising at least the O_2 plasma and an H_2N_2 plasma to the top surfaces of the capacitor switching structures, the surface of the isolation layer.
5. The method of forming the semiconductor structure according to claim 4, wherein the mixed gas plasma has a flow of 100 sccm-15,000 sccm, and a pressure of 10 mtorr-10,000 mtorr.
6. The method of forming the semiconductor structure according to claim 4, wherein the transmitting a mixed gas plasma comprising at least the O_2 plasma and an H_2N_2 plasma to the top surfaces of the capacitor switching structures, the surface of the isolation layer comprises: ionizing, with a radio frequency power of 100 W-10,000 W, a mixed gas comprising O_2 and H_2N_2 , and forming the mixed gas plasma; and transmitting the mixed gas plasma to the top surfaces of the capacitor switching structures, the surface of the isolation layer.
7. The method of forming the semiconductor structure according to claim 1, wherein the removing the oxide layer comprises: removing the oxide layer with a wet etching process.
8. The method of forming the semiconductor structure according to claim 7, wherein the capacitor switching structures are made of a metal material, and the removing the oxide layer with a wet etching process comprises: removing the oxide layer with an alkaline solution as a wet etchant.
9. The method of forming the semiconductor structure according to claim 8, wherein the removing the oxide layer with an alkaline solution as a wet etchant comprises: removing the oxide layer with a mixed solution of NH_4OH and H_2O as the wet etchant.
10. The method of forming the semiconductor structure according to claim 9, wherein a volume ratio of the NH_4OH to the H_2O in the wet etchant is (5:1)-(1,000:1).
11. The method of forming the semiconductor structure according to claim 1, after the removing the oxide layer, further comprising: drying the capacitor base.

12. The method of forming the semiconductor structure according to claim 11, wherein the drying the capacitor base comprises: purging the capacitor base with a mixed gas of nitrogen and isopropanol.

13. A method of forming a semiconductor structure, comprising: forming a capacitor base, the capacitor base comprising a plurality of capacitor switching structures and an isolation layer located between adjacent capacitor switching structures and covering top surfaces of the capacitor switching structures, wherein the isolation layer is made of a nitride material; etching the isolation layer with a mixed gas of CF_4 , CHF_3 and O_2 as an etching gas to exposes the top surfaces of the capacitor switching structures, wherein the top surfaces of the capacitor switching structures are flush with a top surface of the isolation layer, and particulate matters are adhered on the top surfaces of the capacitor switching structures and the top surface of the isolation layer during etching the isolation layer; oxidizing, with an O_2 plasma, the top surfaces of the capacitor switching structures, the top surface of the isolation layer and the particulate matters to form an oxide layer on the top surfaces of the capacitor switching structures and the top surface of the isolation layer; and removing the oxide layer to expose the top surfaces of the capacitor switching structures.

14. The method of forming the semiconductor structure according to claim 13, wherein the forming a capacitor base comprises: providing a substrate, a plurality of capacitor contact regions being provided in the substrate; forming the plurality of the capacitor switching structures on a surface of the substrate, the plurality of the capacitor switching structures being electrically connected to the plurality of the capacitor contact regions in one-to-one correspondence; and forming the isolation layer filling gaps between the adjacent capacitor switching structures and covering the top surfaces of the capacitor switching structures.

15. The method of forming the semiconductor structure according to claim 13, wherein the removing the oxide layer comprises: removing the oxide layer with a wet etching process.

16. The method of forming the semiconductor structure according to claim 15, wherein the capacitor switching structures are made of a metal material, and the removing the oxide layer with a wet etching process comprises: removing the oxide layer with an alkaline solution as a wet etchant.

17. The method of forming the semiconductor structure according to claim 16, wherein the removing the oxide layer with an alkaline solution as a wet etchant comprises: removing the oxide layer with a mixed solution of NH_4OH and H_2O as the wet etchant.

18. The method of forming the semiconductor structure according to claim 17, wherein a volume ratio of the NH_4OH to the H_2O in the wet etchant is (5:1)-(1,000:1).

19. The method of forming the semiconductor structure according to claim 13, after the removing the oxide layer, further comprising: drying the capacitor base.

20. The method of forming the semiconductor structure according to claim 19, wherein the drying the capacitor base comprises: purging the capacitor base with a mixed gas of nitrogen and isopropanol.
